

Multi-Cloud Disaster Recovery System

Technology Stack Document

Tools, Services, and Platforms Used

Document Type: Tech Stack

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Industry Skills: AWS - Azure - Disaster Recovery - Automation - High Availability

1. Overview

This document lists the recommended technology stack for building the Multi-Cloud Disaster Recovery System, organized by architectural layer. Where possible, an AWS-native option, an Azure-native option, and a cloud-agnostic alternative are listed so students/teams can choose based on available accounts and project constraints.

2. Cloud Providers

Provider	Role in Project
Amazon Web Services (AWS)	Primary region - compute, database, storage, monitoring, orchestration
Microsoft Azure	Secondary/standby region - mirrored infrastructure and services

3. Compute & Hosting

Layer	AWS	Azure	Notes
Virtual Machines	EC2	Virtual Machines	Base compute for app servers
Containers	ECS / EKS (Fargate optional)	AKS / Container Apps	Preferred for consistent, portable deployments
Auto Scaling	EC2 Auto Scaling Groups	Virtual Machine Scale Sets	Standby scales from minimal to full capacity on failover
Serverless Functions	AWS Lambda	Azure Functions	Used for orchestration logic and lightweight automation

4. Networking & Traffic Management

Component	AWS	Azure
DNS / Global Traffic Routing	Route 53 (health-check based failover routing)	Azure Traffic Manager / Front Door
Load Balancing	Application Load Balancer (ALB)	Application Gateway
Cross-region/cloud connectivity	Site-to-Site VPN / Direct Connect	VPN Gateway / ExpressRoute
Content Delivery	Amazon CloudFront	Azure CDN

5. Database & Storage

Layer	AWS	Azure	Notes
Relational Database	Amazon RDS / Aurora Global Database	Azure SQL Database (auto-failover groups)	Aurora Global DB gives sub-second cross-region replication
Cross-cloud alt.	PostgreSQL + logical replication / Debezium CDC	Same	Used when a fully vendor-agnostic DB layer is required
Object Storage	Amazon S3 (+ Cross-Region Replication)	Azure Blob Storage (GRS)	Stores backups, static assets, DR snapshots
Caching	Amazon ElastiCache (Redis)	Azure Cache for Redis	Session/query caching, rebuilt on standby
Backup Service	AWS Backup	Azure Backup	Centralized scheduling, retention, restore testing

6. Monitoring, Health Checks & Alerting

Component	AWS	Azure	Notes
Metrics & Alarms	Amazon CloudWatch	Azure Monitor	Composite health scoring source
Synthetic Monitoring	CloudWatch Synthetics / Route 53 Health Checks	Application Insights Availability Tests	Simulated user transactions
Logging	CloudWatch Logs	Azure Log Analytics	Centralized log aggregation
Notifications	Amazon SNS	Azure Notification Hubs	Fan-out to Slack/Email/SMS/ PagerDuty

7. Automation & Orchestration

Purpose	Tool	Notes
Infrastructure as Code	Terraform (primary), AWS CloudFormation / Azure Bicep (alt.)	Ensures identical, repeatable environments per region

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Failover Workflow Engine	AWS Step Functions + Lambda, Azure Logic Apps + Functions	Executes the detect-confirm-promote-redirect-notify state machine
CI/CD Pipeline	GitHub Actions / AWS CodePipeline / Azure DevOps Pipelines	Deploys identical builds to both regions
Configuration Management	Ansible (optional)	Post-provision configuration consistency
Secrets Management	AWS Secrets Manager, Azure Key Vault	Cross-region secret replication

8. Application Layer (Illustrative)

Layer	Suggested Technology
Backend API	Node.js (Express) / Python (FastAPI) / Java (Spring Boot) - any framework the team is comfortable with
Frontend Dashboard	React.js + a charting library (Recharts/Chart.js) for the health/replication dashboard
API Gateway	Amazon API Gateway / Azure API Management
Containerization	Docker, with images pushed to Amazon ECR / Azure Container Registry

9. Security & Identity

- AWS IAM and Azure RBAC for least-privilege role-based access control, mirrored across clouds.
- AWS KMS and Azure Key Vault for encryption key management (data at rest).
- TLS 1.2+ enforced for all data in transit, including inter-region replication traffic.
- AWS Secrets Manager / Azure Key Vault for credentials and API keys, with automated rotation.

10. Suggested Learning / Certification Alignment

- AWS Certified Solutions Architect - Associate (multi-region, HA, DR patterns)
- AWS Certified SysOps Administrator (monitoring, automation)
- Microsoft Certified: Azure Administrator Associate (AZ-104)
- Microsoft Certified: Azure Solutions Architect Expert (AZ-305)
- HashiCorp Certified: Terraform Associate (Infrastructure as Code)